ON THE NONEXISTENCE OF A CYCLIC $(9423, 4711, 2355)$-DIFFERENCE SET

Abstract. We prove nonexistence of a cyclic difference set with parameters $(9423, 4711, 2355)$. The proof uses the multiplier 2 and the quotient of order 3. After translating a hypothetical difference set so that it is fixed by doubling, the two nonzero quotient classes mod 3 have equal size. The corresponding order-3 character value is therefore an integer, but its square must equal 2356, which is impossible.

1. Introduction

Baumert and Gordon list $(9423, 4711, 2355)$ among the small open cyclic Hadamard cases [1], and Gordon's La Jolla status tables continue to retain 9423 on the residual list [2]. We record a one-page nonexistence proof.

Theorem 1. *The cyclic group C_{9423} admits no $(9423, 4711, 2355)$-difference set.*

Proof. Assume for contradiction that $D \subseteq G = C_{9423}$ is a $(9423, 4711, 2355)$-difference set. Then

$$n = k - \lambda = 4711 - 2355 = 2356.$$

Since 2 divides n and $\gcd(2, 9423) = 1$, the multiplier theorem implies that 2 is a multiplier. Thus there exists $t \in G$ such that

$$2D = D + t.$$

Because

$$\gcd(2 - 1, 9423) = 1,$$

we may translate D and assume from the start that

$$2D = D.$$

Let $H = 3G$, so $|H| = 3141$ and $G/H \cong C_3$. Write

$$x_0 = |D \cap H|, \qquad x_1 = |D \cap (1 + H)|, \qquad x_2 = |D \cap (2 + H)|.$$

The doubling map induces multiplication by 2 on $G/H \cong C_3$, so it swaps the two nonzero classes. Since $2D = D$, we obtain

$$x_1 = x_2.$$

Let ψ be a nontrivial character of G/H, and write $\omega = \psi(1 + H)$, so $\omega^2 + \omega + 1 = 0$. Lift ψ to a nonprincipal character χ of G. Then the standard character identity for difference sets gives

$$|\chi(D)|^2 = n = 2356.$$

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On the other hand,

$$\chi(D) = x_0 + x_1\omega + x_2\omega^2 = x_0 + x_1(\omega + \omega^2) = x_0 - x_1,$$

because $x_1 = x_2$ and $\omega + \omega^2 = -1$. Thus $\chi(D)$ is an integer. Consequently $2356 = |\chi(D)|^2$ would be the square of an integer. But

$$2356 = 4 \cdot 19 \cdot 31$$

is not a perfect square. This contradiction proves the theorem. □

Remark 2. The obstruction is already visible on the index-3 quotient. Once the 2-multiplier normalization is in place, the order-3 character sum becomes integral, while the forced norm 2356 is nonsquare.

References

[1] Leonard D. Baumert and Daniel M. Gordon, *On the existence of cyclic difference sets with small parameters*, Fields Inst. Commun. **41** (2004), 29–40.

[2] Daniel M. Gordon, *The La Jolla Difference Set Repository*, ArasuFest talk slides, August 3, 2019.